

Notes: Barber & Odean (QJE, 2001)

Boys Will Be Boys: Gender, Overconfidence, and Common Stock Investment

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1 Big picture

- **Question.** Do more overconfident investors trade more, and does this excessive trading hurt performance?
- **Empirical strategy.** The paper uses gender as a proxy for overconfidence. Psychology research finds that men are generally more overconfident than women in domains such as finance.
- **Main prediction.** If overconfidence causes excessive trading, men should trade more than women and lose more performance from trading.
- **Data.** The authors study common-stock investments of more than 35,000 households at a large discount brokerage from February 1991 through January 1997.
- **Core trading result.** Men trade substantially more than women. In the full sample, women have mean monthly turnover of 4.40%, while men have 6.41%. That is roughly 45% higher turnover for men.
- **Core performance result.** Trading hurts both groups, but hurts men more. Women lose 0.143% per month in net own-benchmark abnormal returns, while men lose 0.221% per month.
- **Annual interpretation.** The net performance penalty from trading is 1.72 percentage points per year for women and 2.65 percentage points per year for men, a difference of 0.94 percentage points per year.
- **Single investors.** The difference is stronger for single investors. Single men trade 67% more than single women and lose 1.44 percentage points per year more from trading.
- **Risk result.** Men hold riskier portfolios: higher market beta, more small-stock exposure, and more volatile stocks.
- **Risk-adjusted performance.** After adjusting for market and size exposures, women earn higher net abnormal returns than men. The two-factor net-return intercept difference is 9 basis points per month, or about 1.1% per year.
- **Security selection result.** Both men and women buy stocks that subsequently underperform the stocks they sell. The difference in security-selection ability between men and women is not statistically significant.
- **Economic takeaway.** The evidence is consistent with overconfidence models: investors who are more prone to overconfidence trade too much, and trading reduces net returns.

2 Data and measurement

2.1 Brokerage account data

- The sample comes from a large discount brokerage firm.
- The brokerage provided position and trading records for roughly 78,000 households.
- The authors focus on common stock investments from February 1991 through January 1997.
- They exclude mutual funds, American Depository Receipts, warrants, and options.
- Households are included if the gender of the person who opened the household's first account can be identified.
- The main gender sample contains 37,664 households: 8,005 female-headed accounts and 29,659 male-headed accounts.

Interpretation. The object of analysis is household-level common-stock trading. The gender assignment is based on the person who opened the first account, not necessarily every person in the household.

2.2 Demographic data

- Demographics come from Infobase Inc. as of June 1997.
- Variables include marital status, children, age, and income.
- A subset of households supplied self-reported net worth, equity share of net worth, and investment experience.
- Married and single households are analyzed separately to sharpen the gender comparison.

Interpretation. The married-versus-single split matters because a married account may reflect joint household decisions. Single men and single women are a cleaner comparison for gender differences.

2.3 Gross and net portfolio returns

For household h in month t , gross common-stock portfolio return is computed as

$$R_{ht}^{gr} = \sum_{i=1}^{S_{ht}} p_{it} R_{it}, \quad (1)$$

where p_{it} is the beginning-of-month portfolio weight of stock i , R_{it} is the stock's gross monthly return, and S_{ht} is the number of common stocks held.

Net returns subtract transaction costs. For security i in month t , the net return is written as

$$1 + R_{it}^{net} = (1 + R_{it}^{gr}) \frac{1 - c_{it}^s}{1 + c_{i,t-1}^p}, \quad (2)$$

where c_{it}^s is the cost of sales and $c_{i,t-1}^p$ is the cost of purchases. These costs include commissions and estimated bid-ask spread costs.

Interpretation. Gross returns isolate the portfolio decisions before trading costs. Net returns measure the realized economic return after the costs generated by trading.

2.4 Turnover

Monthly turnover is defined as

$$Turnover_{ht} = \frac{1}{2}SellTurnover_{ht} + \frac{1}{2}BuyTurnover_{ht}. \quad (3)$$

Sell turnover is sales value during the month divided by beginning-of-month portfolio value. Buy turnover is purchase value during the prior month divided by beginning-of-month portfolio value.

Interpretation. Turnover is the main behavioral outcome. If overconfidence leads investors to believe too strongly in their private signals, they should trade more, raising turnover.

2.5 Own-benchmark abnormal returns

The benchmark for household h is the return that the household would have earned by holding its beginning-of-year portfolio for the rest of the year. The own-benchmark abnormal return is

$$AR_{ht}^{gr} = R_{ht}^{gr} - R_{ht}^B, \quad AR_{ht}^{net} = R_{ht}^{net} - R_{ht}^B, \quad (4)$$

where R_{ht}^B is the return on the beginning-of-year benchmark portfolio.

Interpretation. This is the cleanest performance measure for the paper's question. It asks whether the trades made during the year improve or hurt performance relative to simply doing nothing.

2.6 Security selection

The paper also asks whether performance differences are due to poor stock-picking rather than excessive trading. For each month, it forms portfolios of stocks purchased in the prior twelve months and stocks sold in the prior twelve months:

$$R_t^{buy} = \sum_i w_{it}^{buy} R_{it}, \quad R_t^{sell} = \sum_i w_{it}^{sell} R_{it}. \quad (5)$$

Interpretation. If men underperform because of worse security selection, the stocks they buy should underperform the stocks they sell by more than the analogous comparison for women. The paper finds that both groups make poor trades, but the gender difference in selection is not statistically significant.

2.7 Risk measures

The paper measures risk using:

- portfolio volatility;
- average volatility of individual stocks held;
- market beta;
- exposure to the size factor SMB .

The two-factor risk model is

$$R_{gt} - R_{ft} = \alpha_g + \beta_g(R_{mt} - R_{ft}) + s_gSMB_t + \varepsilon_{gt}, \quad (6)$$

where g indexes the household group, such as female, male, married female, married male, single female, or single male.

Interpretation. Risk adjustment is needed because men hold riskier portfolios. The paper therefore separates trading-cost losses from differences in market and small-stock exposure.

3 Empirical specifications and interpretation

3.1 Main comparison of means

The simplest test compares mean turnover and own-benchmark abnormal returns across female and male households:

$$\Delta Y = \bar{Y}_{women} - \bar{Y}_{men}. \quad (7)$$

- For turnover, the prediction is $\Delta Y < 0$ because men should trade more.
- For net own-benchmark abnormal returns, the prediction is $\Delta Y > 0$ because men should lose more from trading.

Interpretation. The main results are simple differences, but the paper reports them separately for all, married, and single households.

3.2 Cross-sectional regression specification

The paper estimates household-level cross-sectional regressions of the form

$$Y_h = \alpha + \beta_1 Single_h + \beta_2 Woman_h + \beta_3 (Single_h \times Woman_h) + \gamma' X_h + u_h, \quad (8)$$

where Y_h is one of the following outcomes:

- mean monthly turnover;
- monthly own-benchmark abnormal net return;
- portfolio volatility;
- average individual stock volatility;
- market beta;
- size-factor coefficient.

Controls X_h include age, children, income, and an income dummy.

Interpretation. The coefficient on $Woman_h$ measures the difference between married women and married men. The coefficient on $Single_h \times Woman_h$ tells whether the female-male difference is stronger among single households.

3.3 Risk-adjusted return model

For each group, the paper estimates a time-series regression:

$$R_{gt} - R_{ft} = \alpha_g + \beta_g (R_{mt} - R_{ft}) + s_g SMB_t + \varepsilon_{gt}. \quad (9)$$

- α_g is the risk-adjusted abnormal return.
- β_g measures market exposure.
- s_g measures exposure to small stocks.

Interpretation. Men have higher beta and higher SMB exposure. This explains why raw return comparisons are not enough: a riskier portfolio can earn similar gross returns while still losing more after trading costs.

3.4 Overconfidence interpretation

The theoretical chain is

$$\text{Overconfidence} \Rightarrow \text{overestimation of signal precision} \Rightarrow \text{too much trading} \Rightarrow \text{lower net returns.} \quad (10)$$

The gender proxy implies

$$Men > Women \quad \text{in overconfidence,} \quad (11)$$

so the empirical predictions are

$$Turnover_{men} > Turnover_{women}, \quad NetLoss_{men} > NetLoss_{women}. \quad (12)$$

Interpretation. The paper is not claiming that gender directly causes returns. Gender is used as a proxy for overconfidence because psychological evidence suggests men are more overconfident in finance-related tasks.

4 Identification and empirical design

4.1 What identifies the main result

- The empirical comparison uses a large administrative brokerage sample rather than survey self-reports of trading.
- Gender separates investors into groups expected to differ in overconfidence before observing their trading performance.
- Own-benchmark returns compare each household against its own beginning-of-year portfolio, reducing the need to impose a single asset-pricing model.
- Separate results for single investors help address the concern that married households reflect joint decisions.

Interpretation. The key identifying idea is a predicted behavioral difference: if overconfidence drives excessive trading, the group expected to be more overconfident should trade more and lose more from trading.

4.2 Main alternative explanations

- **Risk tolerance.** Men may simply prefer riskier portfolios. The paper addresses this by estimating risk exposures and risk-adjusted returns.
- **Income or wealth.** Men and women may differ in financial capacity. The paper uses demographic controls and reports descriptive statistics.
- **Investment experience.** Men report more investment experience, but the results show that experience does not translate into better net performance.
- **Security selection.** Men might pick worse stocks. The buy-versus-sell analysis shows both men and women pick poorly, with no significant gender difference in selection losses.
- **Joint household decisions.** Married accounts may not represent the account opener alone. Single-household results are stronger, not weaker.

Interpretation. The paper's strongest evidence is the combination of high turnover and worse net own-benchmark returns, especially among singles.

4.3 What the paper cannot fully prove

- Gender is a proxy for overconfidence, not a direct measurement of confidence for every household.
- The data come from one discount brokerage, so the sample may not represent all investors.
- The gender of the account opener may not perfectly identify the person making every trading decision.
- The design is observational and cannot randomly assign overconfidence.
- Other psychological or preference differences may also contribute to trading differences.

Interpretation. The evidence is not a randomized experiment. Its force comes from a sharp theory prediction, a large administrative data set, and consistent patterns in turnover, own-benchmark returns, and risk-adjusted returns.

5 Tables and figures

The paper has no standalone figures. The compact notes therefore group the original tables directly cropped from the paper.

5.1 Table I: Descriptive statistics for female and male households

Read:

- The identified-gender sample contains 8,005 female-headed households and 29,659 male-headed households.
- Women are less likely to be married: 68.0% of women versus 75.7% of men.
- Women are less likely to have children: 25.2% versus 32.2%.
- Mean ages are very similar: 50.9 for women and 50.3 for men.
- Mean incomes are close: \$73,000 for women and \$75,600 for men.
- Men report more investment experience: 48.5% of men versus 39.1% of women report “good” experience, and 14.0% of men versus 8.7% of women report “extensive” experience.

Economic message. The demographic differences are real but not so large that they mechanically explain the main trading result. Men report more experience, yet later tables show that they trade more and lose more from trading.

TABLE I
DESCRIPTIVE STATISTICS FOR DEMOGRAPHICS OF FEMALE AND MALE HOUSEHOLDS

Variable	All households			Married households			Single households		
	Women	Men	Difference (women—men)	Women	Men	Difference (women—men)	Women	Men	Difference (women—men)
Panel A: Infobase data									
Number of households	8,005	29,659	NA	4,894	19,741	NA	2,306	6,326	NA
Percentage married	68.0	75.7	-7.7						
Percentage with children	25.2	32.2	-7.0	33.6	40.4	-6.8	10.6	10.5	0.1
Mean age	50.9	50.3	0.6	49.9	51.1	-1.2	53.0	48.2	4.8
Median age	48.0	48.0	0.0	48.0	48.0	0.0	50.0	46.0	4.0
Mean income (\$000)	73.0	75.6	-2.6	81.2	79.6	1.6	56.7	62.8	-6.1
% with income > \$125,000	11.2	11.7	-0.5	14.2	13.0	1.2	5.9	7.4	-1.5
Panel B: Self-reported data									
Number of households	2,637	11,226		1,707	7,700		652	2,184	
Net worth (\$000)	500.0	500.0	0.0	500.0	500.0	0.0	350.0	450.0	-100.0
50th Percentile	200.0	250.0	-50.0	250.0	250.0	0.0	175.0	200.0	-25.0
Median	100.0	100.0	0.0	100.0	100.0	0.0	100.0	100.0	0.0
25th Percentile	60.0	74.5	-14.5	62.5	74.5	-12.0	40.0	62.0	-22.0
10th Percentile	27.0	37.0	-10.0	35.0	37.0	-2.0	20.0	35.0	-15.0
Equity to net worth (%)									
Mean	13.3	13.2	0.1	12.9	12.9	0.0	14.4	14.3	0.1
Median	6.7	6.7	0.0	6.3	6.6	-0.3	7.9	7.4	0.5
Investment experience (%)									
None	5.4	3.4	2.0	4.7	3.4	1.3	7.4	3.0	4.4
Limited	46.8	34.1	12.7	44.9	34.2	10.7	52.6	33.3	19.3
Good	39.1	48.5	-9.4	40.8	48.5	-7.7	33.3	48.8	-15.5
Extensive	8.7	14.0	-5.3	9.6	13.9	-4.3	6.7	14.9	-8.2

The sample consists of households with common stock investment at a large discount brokerage firm for which we are able to identify the gender of the person who opened the household's first account. Data on marital status, children, age, and income are from Infobase Inc. as of June 1997. Self-reported data are information supplied to the discount brokerage firm at the time the account is opened by the person on opening the account. Income is reported within eight ranges, where the top range is greater than \$125,000. We calculate means using the subject of each range and \$125,000 for the top range. Equity to Net Worth (%) is the proportion of the market value of common stock investment at this discount brokerage firm as of January 1991 to total self-reported net worth when the household opened its first account at this brokerage. Those households with a proportion equity to net worth greater than 100 percent are deleted when calculating means and medians. Number of observations for each variable is slightly less than the number of reported

TABLE II
POSITION VALUE, TURNOVER, AND RETURN PERFORMANCE OF COMMON STOCK INVESTMENTS OF FEMALE AND MALE HOUSEHOLDS: FEBRUARY 1991 TO JANUARY 1997

	All households			Married households			Single households		
	Women	Men	Difference (women—men)	Women	Men	Difference (women—men)	Women	Men	Difference (women—men)
Number of households	8,005	29,659	NA	4,894	19,741	NA	2,306	6,326	NA
Panel A: Position Value and Turnover									
Mean [median] beginning position value (\$)	18,371 [7,387]	21,975 [8,218]	-3,604*** [-831]***	17,754 [7,410]	22,293 [8,175]	-4,539*** [-1,765]***	19,654 [7,491]	20,161 [8,097]	-507*** [-606]***
Mean [median] monthly turnover (%)	4.40 [1.74]	6.41 [2.94]	-2.01*** [-1.20]***	4.41 [1.79]	6.11 [2.81]	-1.70*** [-1.02]***	4.22 [1.55]	7.05 [3.32]	-2.83*** [-1.77]***
Panel B: Performance									
Own-benchmark monthly abnormal gross return (%)	-0.041*** (-2.84)	-0.069*** (-3.66)	0.028*** (2.43)	-0.050*** (-2.89)	-0.068*** (-3.67)	0.018 (1.28)	-0.029* (-1.64)	-0.074*** (-3.60)	0.045*** (2.53)
Own-benchmark monthly abnormal net return (%)	-0.143*** (-9.70)	-0.221*** (-10.83)	0.078*** (6.35)	-0.154*** (-9.10)	-0.214*** (-10.48)	0.060*** (3.95)	-0.121*** (-6.68)	-0.242*** (-11.15)	0.120*** (6.68)

***, **, * indicate significant at the 1, 5, and 10 percent level, respectively. Tests for differences in medians are based on a Wilcoxon sign-rank test statistic. Households are classified as female or male based on the gender of the person who opened the account. Beginning position value is the market value of common stocks held in the first month that the household appears during our sample period. Mean monthly turnover is the average of sales and purchase turnover. (Median values are in brackets.) Own-benchmark abnormal returns are the average household percentage monthly abnormal returns calculated as the realized monthly return for a household less the return that would have been earned if the household had held the beginning-of-year portfolio for the entire year (i.e., the twelve months beginning February 1). T-statistics for abnormal returns are in parentheses and are calculated using time-series standard errors across months.

5.2 Table II: Turnover and own-benchmark performance

Read:

- Women have mean monthly turnover of 4.40%; men have 6.41%. Men therefore trade about 45% more than women.
- The difference is larger among singles: single women have turnover of 4.22%, while single men have 7.05%.
- Gross own-benchmark abnormal returns are negative for both groups: -0.041% per month for women and -0.069% for men.
- Net own-benchmark abnormal returns are more negative: -0.143% per month for women and -0.221% for men.
- The female-minus-male difference in net own-benchmark return is 0.078% per month for all households and 0.120% per month for single households.

Economic message. The table is the core result. Both men and women lose money by trading relative to doing nothing, but men trade more and therefore lose more.

5.3 Tables III and IV: Regression controls and risk adjustment

Read: Table III:

- The cross-sectional turnover regression gives a woman coefficient of -1.461 percentage points per month.
- The interaction of single and woman is -0.733, implying an even larger gender turnover gap among singles.
- The own-benchmark net return regression gives a positive woman coefficient of 0.058 and a positive single-woman interaction of 0.027.
- Men have higher risk exposures: the woman coefficient is negative for portfolio volatility, individual stock volatility, beta, and the size coefficient.

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TABLE III
CROSS-SECTIONAL REGRESSIONS OF TURNOVER, OWN-BENCHMARK ABNORMAL
RETURN, BETA, AND SIZE: FEBRUARY 1991 TO JANUARY 1997

Dependent variable	Mean monthly turnover (%)	Own-benchmark abnormal net return	Portfolio volatility	Individual volatility	Beta	Size coefficient
Intercept	6.269*** (-11.47)	-0.321*** (-11.47)	11.466*** (58.85)	11.658*** (70.98)	1.226+ (11.44)	0.776*** (22.16)
Single	0.483*** (4.24)	0.002 (0.14)	0.320*** (3.40)	0.330*** (4.17)	0.020** (2.12)	0.079*** (4.65)
Woman	-1.461*** (-12.76)	0.058*** (4.27)	-0.689*** (-7.27)	-0.682*** (-8.54)	-0.037*** (-3.91)	-0.136*** (-8.00)
Single × woman	-0.733*** (-3.38)	0.027 (1.08)	-0.439** (-2.45)	-0.540*** (-3.57)	-0.029 (-1.60)	-0.138*** (-4.30)
Age/10	-0.311*** (-9.26)	0.002*** (4.23)	-0.536*** (-19.31)	-0.393*** (-16.78)	-0.027*** (-9.55)	-0.055*** (-11.00)
Children	-0.037 (-0.40)	0.008 (0.76)	-0.014 (-0.19)	-0.051 (-0.79)	-0.002 (-0.22)	-0.008 (-0.61)
Income /1000	-0.002 (-1.30)	0.0002 (1.33)	0.0003 (0.22)	0.001 (1.38)	0.003 (2.49)**	0.001 (0.31)
Income dummy	-0.0003 (-0.24)	0.027 (1.54)	0.011 (0.10)	0.012 (0.11)	-0.008 (-0.68)	-0.018 (-0.82)
Adj. R ² (%)	1.53	0.20	2.11	1.95	0.59	1.19

TABLE IV
RISK EXPOSURES AND RISK-ADJUSTED RETURNS OF COMMON STOCK INVESTMENTS OF FEMALE
AND MALE HOUSEHOLDS: FEBRUARY 1991 TO JANUARY 1997

	All households			Married households			Single households		
	Women	Men	Difference (women-men)	Women	Men	Difference (women-men)	Women	Men	Difference (women-men)
Number of households	8,005	29,659	NA	4,894	19,741	NA	2,306	6,326	NA
Panel A: Gross average household percentage monthly returns									
Two-factor model intercept	-0.044	-0.083	0.039	-0.051	-0.082	0.031	-0.036	-0.099	0.063
Two-factor model coefficient estimate on $(R_{mt} - R_{ft})$	1.050***	1.081***	-0.031**	1.053***	1.075***	0.022*	1.035***	1.088***	0.053***
Two-factor model coefficient estimate on SMB_t	0.360***	0.519***	-0.159***	0.380***	0.490***	0.109***	0.307***	0.582***	0.275***
Adjusted R ²	93.8	92.0	65.3	93.4	92.1	52.8	94.4	91.5	0.275***
Panel B: Net average household percentage monthly returns									
Two-factor model intercept	-0.162*	-0.253**	0.091***	-0.171**	-0.245**	0.074**	-0.142**	-0.285**	0.143***

***, **, * indicate significant at the 1, 5, and 10 percent level, respectively.
Households are classified as female or male based on the gender of the person who opened the account. Households are classified as married or single based on the marital status of the head of household. Coefficient and intercept estimates for the two-factor model are those from a time-series regression of the gross (net) average household excess returns on the market excess return $(R_{mt} - R_{ft})$ and a zero-investment size portfolio (SMB_t) : $(R_{it} - R_{ft}) = \alpha_i + \beta_1(R_{mt} - R_{ft}) + \beta_2SMB_t + \epsilon_{it}$.

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Read: Table IV:

- Men have higher market betas and higher size-factor exposure than women.
- For all households, the net two-factor intercept is -0.162% for women and -0.253% for men.
- The female-minus-male net intercept difference is 0.091% per month, or about 1.1% per year.
- For singles, the net intercept difference is 0.143% per month, again larger than the all-household estimate.

Economic message. These tables address the main risk-based alternative explanation. Men hold riskier portfolios, but after accounting for market and size exposures, women still have better net performance.

6 Short summary

- The paper tests overconfidence models of excessive trading using gender as a proxy for overconfidence.
- Psychology research predicts that men should be more overconfident than women in financial tasks.
- The data cover common-stock trades and positions of more than 35,000 households at a discount brokerage from 1991 to 1997.
- Men trade about 45% more than women in the full sample.
- Single men trade about 67% more than single women.
- Trading lowers net returns for both men and women.
- The net trading penalty is larger for men: 2.65 percentage points per year versus 1.72 percentage points for women.
- The male-female difference is stronger among single households.
- Men hold riskier portfolios, with higher market beta, more small-stock exposure, and higher volatility.
- Risk-adjusted return tests still show that women outperform men on net returns.
- The buy-versus-sell tests show that both men and women make poor trades; men do not simply have uniquely worse stock-picking ability.
- The paper concludes that overconfidence is a plausible explanation for excessive trading by individual investors.

7 Conclusion

7.1 Core conclusion

Barber and Odean show that the group expected to be more overconfident trades more and loses more from trading. Men trade more frequently than women, and this higher turnover lowers their net investment performance. The strongest evidence comes from single men and single women, where gender-based differences in trading and returns are largest.

7.2 Economic intuition

Overconfident investors overestimate the precision of their information. They therefore believe their trades are more valuable than they really are. Trading is costly because of commissions, bid-ask spreads, and poor timing or selection. The result is excessive turnover and lower net returns. Gender provides a useful empirical partition because psychological evidence predicts different levels of overconfidence across men and women.

7.3 Takeaway

The paper is a classic behavioral finance test. It links a psychological bias to a direct market behavior and to realized investment performance. The main message is not simply that men trade more than women. It is that excessive confidence can transform private beliefs into costly trading.